MENOUFIA UNIVERSITY

SYSTEM ANALYSIS & DESIGN PROJECT

SECOND YEAR

2024-2025

جامعة المنوفية

كلية الحاسبات والمعلومات

2024-2025

Public Facility Reservation System

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Planning (Project Plan)

Project Identification

Project Name:

Public Facility Reservation System

Current Problem:

1- Citizens must call or visit the facility administration office to reserve a hall, sports court, or meeting room, which causes long waiting and busy phone lines.
2- Reservations are written on paper or in notebooks, so double bookings and lost reservations happen.
3- Citizens cannot know whether a facility is available, or the status of their request (pending, approved, rejected, cancelled), without calling or visiting the office.
4- There is no digital booking history for each facility or user, and no reminders before the reservation date.
5- Management cannot easily follow facility usage, schedules, and maintenance closures, or produce usage reports.

Description about project:

- The system is a web application (also usable from mobile phones) that lets citizens register and reserve public facilities (community halls, sports courts, meeting rooms, parks) online by choosing the facility, the date and an available time slot. After the request is submitted, the facility administrator reviews it and updates its status (Pending, Approved, Rejected, Cancelled, Completed) and the citizen is notified automatically by SMS / e-mail / in-app notification at every change. The administrator manages facilities, schedules, fees and maintenance closures, and the system keeps the full reservation history of every facility and user and produces daily and monthly usage reports.

System Request

System Request—Public Facility Reservation System	
Project Sponsor:	Facilities Administration Manager (owner of the public facilities)
Business Need:	1- Reduce waiting time and phone calls for reservations. 2- Avoid double bookings and lost reservations. 3- Keep citizens informed about their reservation status at all times. 4- Make reservation available 24/7 for everyone.
Business Requirements:	1- Online reservation with available time slots. 2- Reservation approval and status updates by the staff (tablet / PC). 3- Automatic notifications (SMS / e-mail). 4- Admin dashboard, reservation history and reports.
Business Value:	1- Saving time (staff and customers). 2- Saving cost (less paper, fewer no-shows). 3- Higher citizen satisfaction and trust. 4- More reservations and better use of facility capacity.
Special Issues or Constraints:	1- Deadline: the delivery date set by the instructor (end of semester). 2- Limited budget (about 72,000 EGP one-time) and student-team skills. 3- Customers' personal data must be protected.

Feasibility Analysis

Technical Feasibility

1. The system will be a responsive web application: HTML, CSS and JavaScript for the front end, a back-end framework (e.g. Node.js or PHP/Laravel) and a MySQL / PostgreSQL database. These are common technologies that the team already studied.
2. Development tools needed: Visual Studio Code, Git/GitHub for teamwork, and a database tool such as MySQL Workbench or pgAdmin.
3. The system needs an internet connection and cloud hosting so citizens can check availability and reserve online, and an SMS / e-mail gateway to send notifications.
4. It runs in any modern browser on Windows, Android or iOS, and staff can review and update reservations from a tablet or PC. Conclusion: technically feasible.

Economic Feasibility

1. One-time cost is about 72,000 EGP (development 40,000, design 8,000, hosting setup 3,000, testing 5,000, training 4,000, tablets / entrance kiosks 12,000).
2. Yearly operating cost is about 21,760 EGP (hosting 6,000, SMS 5,760, maintenance 10,000).
3. Expected yearly benefits are about 95,760 EGP at full use (staff time saved 24,960, fewer no-shows 36,000, extra reservations and fee revenue 28,800, paper savings 6,000).
4. Payback period is about 12 months and the 3-year ROI is about 208%, so the project is economically feasible. Details and formulas are in the attached Excel file (all amounts are team estimates).

Organizational Feasibility

Organizational Feasibility Analysis includes the assessment of:

1. Management prowess: the facilities administration manager sponsors the project and can approve new reservation procedures; the student team has a clear plan with roles and deadlines under the instructor's supervision.
2. Resource sufficiency: human resources (developers, receptionists and facility staff who will use the system), facilities (2–3 tablets / PCs at the administration office and facility entrances, internet connection), and intellectual property (source code kept in a private GitHub repository, citizen data protected by login and roles).
3. Expected resistance: staff used to paper may resist change; this is handled by short training sessions, a simple interface and a phased start (online reservation first, then status updates and reports). Conclusion: organizationally feasible.